

School of Natural Sciences and Mathematics

Physics (BA, BS)

The science of physics seeks understanding of the behavior of matter and energy at the most general and fundamental level. The physicist is trained to explore the physical universe in which people live and seeks interpretations of the natural phenomena found there. While much is known about the physical universe, many phenomena still remain to be investigated, understood, and exploited to the ultimate benefit of humankind. This is the challenge that a modern physicist faces.

The Degrees

The student majoring in Physics must meet the general University requirements for admission and for the specific degree the student is seeking. The Physics Program offers both the Bachelor of Arts and the Bachelor of Science degrees. A total of 120 semester credit hours is required for either degree. With the proper sequencing of courses, these degrees can be achieved in a four year period.

Bachelor of Arts

The [Bachelor of Arts](#) program provides an opportunity for a strong base in physics for students wishing to pursue graduate studies (non-physics) in, for example, business administration, economics, finance, oceanography, and patent or high technology law. Additionally, students seeking certification as high school teachers with physics as a major specialization and those seeking employment in industry, government service, and computer technology have the opportunity to obtain the necessary physics background through the BA program. The lower-division course requirements for the BA degree are the same as those for the BS degree. At the upper-division level, the physics core course are identical with the BS core courses, but the physics elective is replaced with [PHYS 4398](#) Senior Research Project for BA Degree and 15 semester credit hours of advanced physics courses are replaced with 15 semester credit hours of upper-division science electives.

Bachelor of Science

The [Bachelor of Science](#) is intended for students interested in a professional career in physics or closely related fields. It provides an excellent background for graduate programs in physics, biophysics, geophysics, engineering, medicine and other health related degree programs.

Graduate Studies Track

The recommended course of study toward a Bachelor of Science degree for those students who intend to pursue graduate studies in Physics begins with a two-semester Honors sequence of fundamentals of physics that gives the student a more extensive foundation in basic physics. The remainder of the program is the same as the regular BS program.

Algebra Based Physics

An algebra based general physics course ([PHYS 1301](#), [PHYS 1302](#)) with lab ([PHYS 2125](#), [PHYS 2126](#)) is offered for students interested in the health sciences and those curious about the physical world in which we live. It stresses understanding the workings of nature and the physical processes and phenomena occurring therein.

Bachelor of Arts in Physics

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

FACG> nsm-physics-ba

Professors: Phillip C. Anderson, Lunjin Chen, Gerardo Cisneros, Shengwang Du, Matthew J. Goeckner, Mustapha Ishak-Boushaki, Michael Kesden, David J. Lary, Mark Lee, Roger Malina, Anton V. Malko, Jason D. Slinker, Robert M. Wallace, Anvar A. Zakhidov, Bei Zeng, Chuanwei Zhang, Fan Zhang

Associate Professors: Fabiano Da Silveira Rodrigues, Yuri Gartstein, Lindsay J. King, Michael Kolodrubetz, Lloyd Lumata, Bing Lv, Kaloyan Penev, Xiaojia Zhang

Assistant Professors: Zihao Ou, Aaron Smith, Ivan Vasko, Qingyu Zhu

Professors Emeriti: Austin J. Cunningham, Robert Glosser, Joseph M. Izen, Myron B. Salamon, Brian A. Tinsley

Research Professor: Roderick A. Heelis

Professor of Instruction: Paul Mac Alevey

Associate Professors of Instruction: Amena Khan, Lamya Saleh

Assistant Professor of Instruction: Roger Kadala

UT Dallas Affiliated Faculty: Kyeongjae (KJ) Cho, Massimo V. Fischetti, Heather Hayenga, Julia W. P. Hsu, Stephen D. Levene, Lawrence J. Overzet, A. Dean Sherry, Mary L. Urquhart

Professor Emeritus: Xinchou Lou

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2413](#) Differential Calculus^{1,2}

or [MATH 2417](#) Calculus I^{1,2}

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[PHYS 2325](#) Mechanics³

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{3,4}

[PHYS 2326](#) Electromagnetism and Waves³

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves^{3,4}

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[CHEM 1311](#) General Chemistry I₃

or [CHEM 1315](#) Honors Freshman Chemistry I₃

[CHEM 1312](#) General Chemistry II₃

or [CHEM 1316](#) Honors Freshman Chemistry II₃

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 67-69 semester credit hours

Major Preparatory Courses: 25-27 semester credit hours beyond core curriculum

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I₃

or [CHEM 1315](#) Honors Freshman Chemistry I₃

[CHEM 1312](#) General Chemistry II₃

or [CHEM 1316](#) Honors Freshman Chemistry II₃

[MATH 2413](#) Differential Calculus_{1,2}

or [MATH 2417](#) Calculus I_{1,2}

[MATH 2414](#) Integral Calculus₂

or [MATH 2419](#) Calculus II₂

[MATH 2415](#) Calculus of Several Variables₂

or [MATH 3351](#) Advanced Calculus₂

[NATS 1101](#) Natural Sciences and Mathematics Freshman Seminar

[MATH 2418](#) Linear Algebra

[MATH 2420](#) Differential Equations with Applications

[PHYS 1100](#) The Fun of Physics

[PHYS 2303](#) Contemporary Physics

[PHYS 2325](#) Mechanics₃

and [PHYS 2125](#) Physics Laboratory I₅

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat_{3 4 6}

[PHYS 2326](#) Electromagnetism and Waves₃

and [PHYS 2126](#) Physics Laboratory II₅

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves_{3 4}

and [PHYS 2126](#) Physics Laboratory II₅

Major Core Courses: 30 semester credit hours

[PHYS 3312](#) Classical Mechanics

[PHYS 3427](#) Electronics with Laboratory

[PHYS 3330](#) Numerical Methods in Physics and Computational Techniques

[PHYS 3411](#) Theoretical Physics

[PHYS 3416](#) Electricity and Magnetism

[PHYS 4311](#) Thermodynamics and Statistical Mechanics

[PHYS 4373](#) Physical Measurements Laboratory

[PHYS 4398](#) Senior Research Project for BA Degree

Choose one Physics Elective course from the following:

[PHYS 3380](#) Astronomy

[PHYS 4301](#) Quantum Mechanics I

[PHYS 4302](#) Quantum Mechanics II

[PHYS 4340](#) Introduction to Quantum Information

[PHYS 4346](#) Quantum Physics for Computing and Information

[PHYS 4352](#) Concepts of Modern Physics

[PHYS 4371](#) Solid State Physics

[PHYS 4386](#) Elementary Particle Physics

[PHYS 4392](#) Extragalactic Astrophysics

[PHYS 4V07](#) Senior Research Projects

[PHYS 4V10](#) Research Topics in Physics

[PHYS 4V11](#) Topics in Physics

Major Related Courses: 12 semester credit hours

12 semester credit hours of upper-division Science Electives

III. Elective Requirements: 9-11 semester credit hours

Both lower- and upper-division courses may count as electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Bachelor of Science in Physics

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

FACG> nsm-physics-bs

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Professor Emeritus: Xinchou Lou

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Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2413](#) Differential Calculus^{1,2}

or [MATH 2417](#) Calculus I^{1,2}

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[PHYS 2325](#) Mechanics³₋

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{3 4}_{-,-}

[PHYS 2326](#) Electromagnetism and Waves³₋

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves^{3 4}_{-,-}

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[CHEM 1311](#) General Chemistry I³₋

or [CHEM 1315](#) Honors Freshman Chemistry I³₋

[CHEM 1312](#) General Chemistry II³₋

or [CHEM 1316](#) Honors Freshman Chemistry II³₋

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 67-69 semester credit hours

Major Preparatory Courses: 25-27 semester credit hours beyond core curriculum

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I₃

or [CHEM 1315](#) Honors Freshman Chemistry I₃

[CHEM 1312](#) General Chemistry II₃

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[PHYS 1100](#) The Fun of Physics

[PHYS 2303](#) Contemporary Physics₅

[PHYS 2325](#) Mechanics₃

and [PHYS 2125](#) Physics Laboratory I₅

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{3 4 6}_{_,_,_}

[PHYS 2326](#) Electromagnetism and Waves³_{_}

and [PHYS 2126](#) Physics Laboratory II⁵_{_}

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves^{3 4}_{_,_}

and [PHYS 2126](#) Physics Laboratory II⁵_{_}

Major Core Courses: 24 semester credit hours

[PHYS 3312](#) Classical Mechanics

[PHYS 3427](#) Electronics with Laboratory

[PHYS 3330](#) Numerical Methods in Physics and Computational Techniques

[PHYS 3411](#) Theoretical Physics

[PHYS 3416](#) Electricity and Magnetism

[PHYS 4311](#) Thermodynamics and Statistical Mechanics

[PHYS 4373](#) Physical Measurements Laboratory

Major Related Courses: 18 semester credit hours

[PHYS 4301](#) Quantum Mechanics I

[PHYS 4302](#) Quantum Mechanics II

[PHYS 4328](#) Optics

[PHYS 4352](#) Concepts of Modern Physics

Choose two Physics Elective courses from the following:

[PHYS 3380](#) Astronomy

[PHYS 4340](#) Introduction to Quantum Information

[PHYS 4371](#) Solid State Physics

[PHYS 4386](#) Elementary Particle Physics

[PHYS 4392](#) Extragalactic Astrophysics

[PHYS 4V07](#) Senior Research Projects

[PHYS 4V10](#) Research Topics in Physics

[PHYS 4V11](#) Topics in Physics

III. Elective Requirements: 9-11 semester credit hours

Both lower- and upper-division courses may count as electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

UTeach Option

The [UTeach option](#) may be added to either the BA or BS degree. UTeach Dallas Option degree plans are streamlined to allow students to complete both a rigorous Bachelor of Science or Bachelor of Arts degree and all coursework for middle or high school teacher certification in four years. Teaching Option degrees require deep content knowledge combined with courses grounded in the latest research on math and science education. While most graduates go on to classroom teaching, UTeach alums are also prepared to enter graduate school and to work in a discipline related industry.

Fast Track Baccalaureate/Master's Degrees

For students with strong academic record who intend to pursue master's studies at UT Dallas, accelerated BA/MS or BS/MS Fast Tracks are available. After Fast Track admission to an MS program, students may take up to 15 semester credit hours of approved graduate courses in their senior year to use toward completion of both the BA/BS and MS degrees. Students must still complete all other requirements for their undergraduate degrees. The available Fast Track options and their admission requirements, if any, in addition to having completed 90 or more semester credit hours (out of which 36 hours are from the core curriculum) with a cumulative GPA of at least 3.00, are as follows:

- [MS in Physics](#): Students must have a cumulative GPA of at least 3.20 in a minimum of 30 semester credit hours of upper-division courses that include [PHYS 3411](#), [PHYS 3312](#), [PHYS 3330](#), [PHYS 3416](#), [PHYS 4301](#), and [PHYS 4311](#).
- [MS in Mathematics](#): (Only Data Science, Applied Mathematics, and Mathematics for Decision and Engineering Sciences specializations) Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.
- [MS in Data Science and Statistics](#): (Only Data Science and Applied Statistics specializations) Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.
- [MS in Bioinformatics and Computational Biology](#): Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.

- [MS in Molecular and Cell Biology](#): Students must have a GPA of at least 3.20 and have completed [BIOL 2311](#), [BIOL 2312](#), and [BIOL 2281](#), or equivalent courses.
- [MS in Biotechnology](#): Students must have a GPA of at least 3.20 and have completed [BIOL 2311](#), [BIOL 2312](#), and [BIOL 2281](#), or equivalent courses.

Interested students, after reviewing the [UT Dallas Fast Track policy](#), should contact their undergraduate advisor and the graduate advisor of the intended MS program well in advance of their junior year to prepare a course sequence permitting maximal advantage and apply to the Fast Track program.

1. One semester credit hour of Calculus is counted as Major Preparatory credit; three semester credit hours are counted in Core Curriculum.
2. Students may choose either calculus sequence MATH 2413, MATH 2414, and MATH 2415 or MATH 2417, MATH 2419 and MATH 3351.
3. A required Major preparatory course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.
4. If Honors Physics is taken, three hours are counted under the sciences core and one hour is counted under Physics Major Preparatory Courses.
5. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
6. Students who complete PHYS 2421 do not need to complete PHYS 2125.

Minors

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

For all minors in the School of Natural Sciences and Mathematics students must complete all prerequisite sequences for required minor courses.

Minor in Physics

20 semester credit hours

Required: 12 semester credit hours

[PHYS 2325](#) Mechanics

[PHYS 2125](#) Physics Laboratory I

[PHYS 2326](#) Electromagnetism and Waves

[PHYS 2126](#) Physics Laboratory II

[PHYS 3411](#) Theoretical Physics

Also:

Three other upper-division physics courses

Certificates

Certificate in Quantum Information

15 semester credit hours

A Certificate in Quantum Information is offered by the Department of Physics in the School of Natural Sciences and Mathematics.

The focus of the Certificate in Quantum Information is to provide science oriented students with the necessary skills and knowledge for job opportunities in the quantum industry.

Admission Requirements

Undergraduate students who have completed two semesters of undergraduate calculus and one semester of undergraduate linear algebra.

Certificate Requirements

Students must complete the following courses:

[PHYS 4346](#) Quantum Physics for Computing and Information

or [PHYS 4301](#) Quantum Mechanics I

[PHYS 3330](#) Numerical Methods in Physics and Computational Techniques⁷

[PHYS 4340](#) Introduction to Quantum Information

[PHYS 4350](#) Quantum Algorithm and Software

[PHYS 4347](#) Quantum Network and Communication

7. Equivalent programming courses, such as CS 1325, CS 1336, BMEN 1300, or MATH 2370, can

substitute for PHYS 3330.

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